



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

QUARKUS PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on native image, CDI, Panache, reactive, and testing

TOTAL: 10 QUESTIONS

Q1 What is Quarkus and why is it designed for cloud-native environments?

Threat Model

Q6 What are Uni and Multi types in SmallRye Mutiny?

Observability

Q2 How does Quarkus native compilation with GraalVM work?

Architecture

Q7 What is Quarkus Panache and how does it simplify JPA?

Lifecycle

Q3 How does Quarkus's build-time CDI differ from standard runtime CDI?

Configuration

Q8 How do you consume Kafka messages in Quarkus?

Compliance

Q4 How do you build a REST endpoint in Quarkus with RESTEasy Reactive?

Hardening

Q9 What happens during Quarkus's augmentation (build) phase?

Testing

Q5 What are Quarkus Dev Services?

SSO / IdP

Q10 How do you test a Quarkus application?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Quarkus is a Kubernetes-native Java framework

Mitigation

Quarkus is a Kubernetes-native Java framework that moves as much work as possible to build time. This produces extremely fast startup (< 50 ms native) and low memory footprint, which directly reduces cloud compute costs and enables efficient scale-to-zero.

A6 Uni<T> represents a single asynchronous

Observability

Uni<T> represents a single asynchronous result (0 or 1 item), similar to a promise. Multi<T> is a reactive stream of multiple items. Both are lazy they start processing only when subscribed. Quarkus RESTEasy Reactive handles subscription automatically.

A2 Quarkus performs AOT analysis at build

Primitives

Quarkus performs AOT analysis at build time using GraalVM native-image. It traces reachable classes/methods, eliminates dead code, and compiles to a platform-specific binary. Reflection and dynamic proxies must be declared via metadata for native builds.

A7 Panache provides two patterns: Active Record

Automation

Panache provides two patterns: Active Record (entities extend PanacheEntity and have static query methods like `User.find('email', email)`) and Repository (inject a `PanacheRepository`). Both reduce boilerplate compared to raw JPA repositories.

A3 Quarkus resolves CDI injection points, validates

Hardening

Quarkus resolves CDI injection points, validates beans, and generates synthetic bytecode at build time using the Jandex index. This eliminates the runtime reflection that standard CDI requires, reducing startup time and memory significantly.

A8 Add

Governance

Add `quarkus-smallrye-reactive-messaging-kafka`. Annotate a method with `@Incoming('topic-name')` and define kafka channel configuration in `application.properties`. The framework handles consumer lifecycle, deserialization, and back-pressure automatically.

A4 Annotate a class with `@Path('/items')` and

Gotchas

Annotate a class with `@Path('/items')` and methods with `@GET`, `@POST` using standard JAX-RS annotations. RESTEasy Reactive processes requests on the event loop without blocking; return `Uni<T>` from SmallRye Mutiny for true reactive endpoints.

A9 Augmentation runs annotation processors,

Assessment

Augmentation runs annotation processors, resolves CDI, validates configuration, generates synthetic classes, and produces a pre-built bytecode store. At runtime, the app loads pre-computed data instead of re-analyzing classpath, giving near-instant startup.

A5 Dev Services automatically start required

Federation

Dev Services automatically start required infrastructure (PostgreSQL, Kafka, Redis) as Testcontainers when running in dev or test mode and no explicit connection config is found. This eliminates the need for local service setup during development.

A10 `@QuarkusTest` starts the application on a

Optimization

`@QuarkusTest` starts the application on a random port for the test. Inject a REST-assured spec to call endpoints. `@QuarkusIntegrationTest` tests the packaged artifact (JVM JAR or native binary). Use `@InjectMock` to replace CDI beans with Mockito mocks.